Supplementary Information

for: The Energy Crossover Effect: How Model Architecture Shapes Hardware Efficiency Across Generative AI Modalities

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1 Supplementary Figures

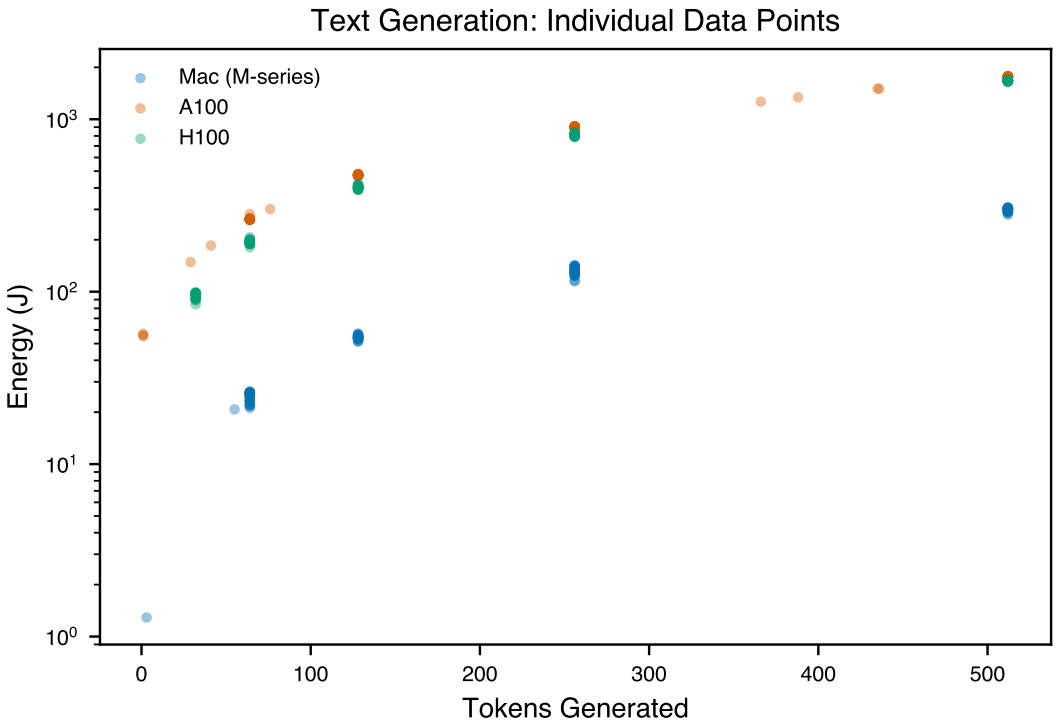

Figure 1: Text generation: individual data points. All individual energy measurements for text generation using Phi-3-mini-4k across three hardware platforms (Mac M-series, NVIDIA A100, NVIDIA H100). Each point represents a single inference run. The Mac platform shows substantially lower absolute energy consumption at all token counts, while GPU platforms exhibit higher but more consistent energy draw due to high idle power.

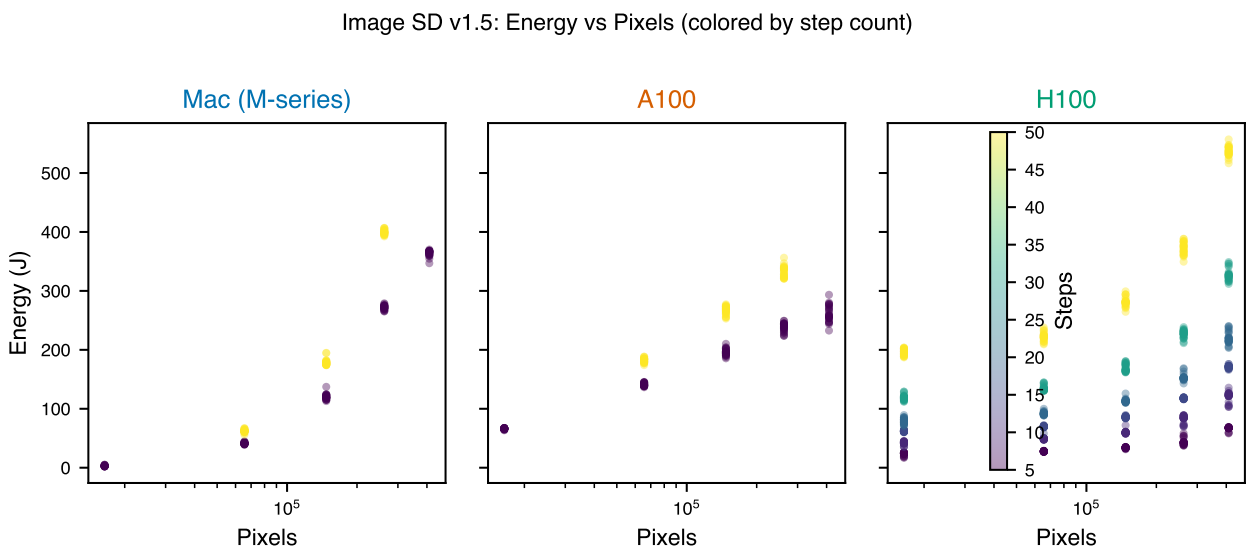

Figure 2: Image generation (SD v1.5): individual data points colored by step count. Energy consumption for Stable Diffusion v1.5 image generation across all three platforms. Points are colored by the number of diffusion steps. Higher step counts and larger resolutions both increase energy consumption, but the scaling behavior differs markedly between platforms.

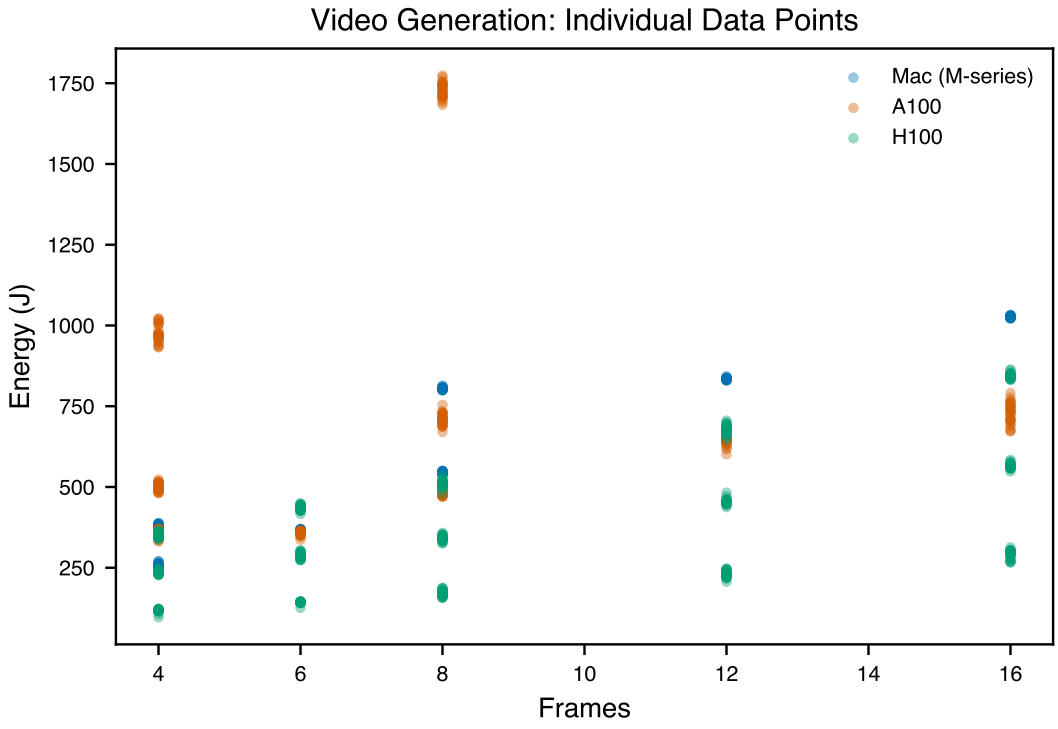

Figure 3: Video generation: individual data points. All individual energy measurements for AnimateDiff video generation across available platforms. Energy scales with frame count, with the crossover between on-device and cloud efficiency depending on the number of frames generated.

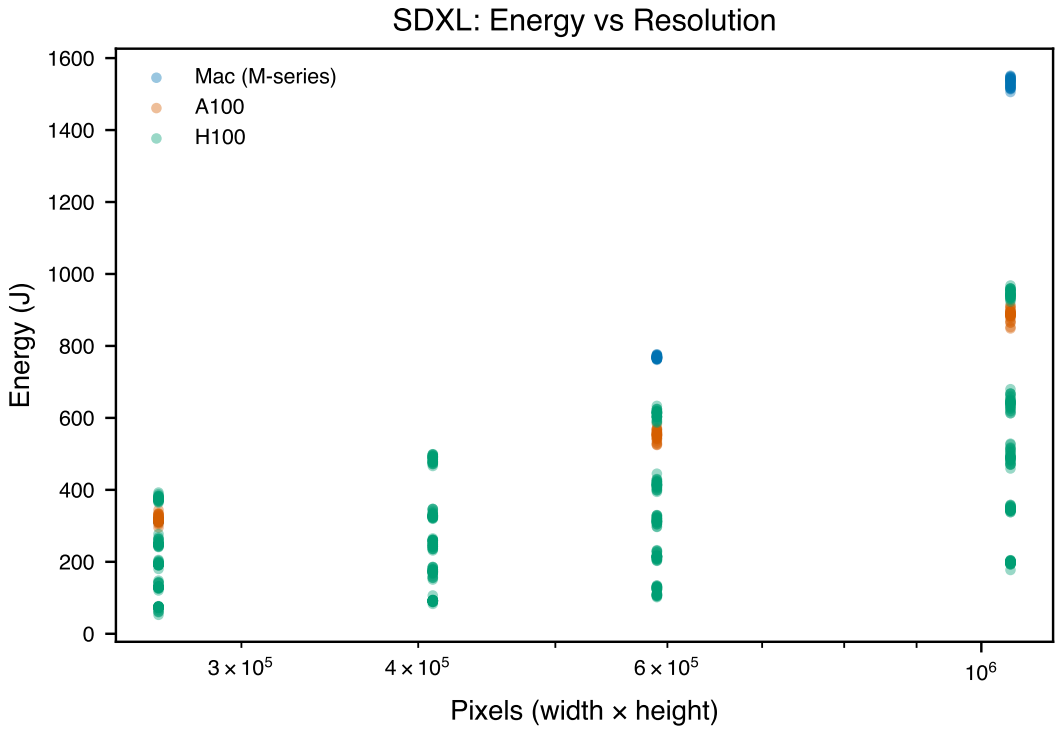

Figure 4: SDXL cross-hardware comparison. Energy consumption for SDXL image generation across three platforms as a function of output resolution. SDXL requires substantially more compute than SD v1.5, shifting the crossover point to lower resolutions.

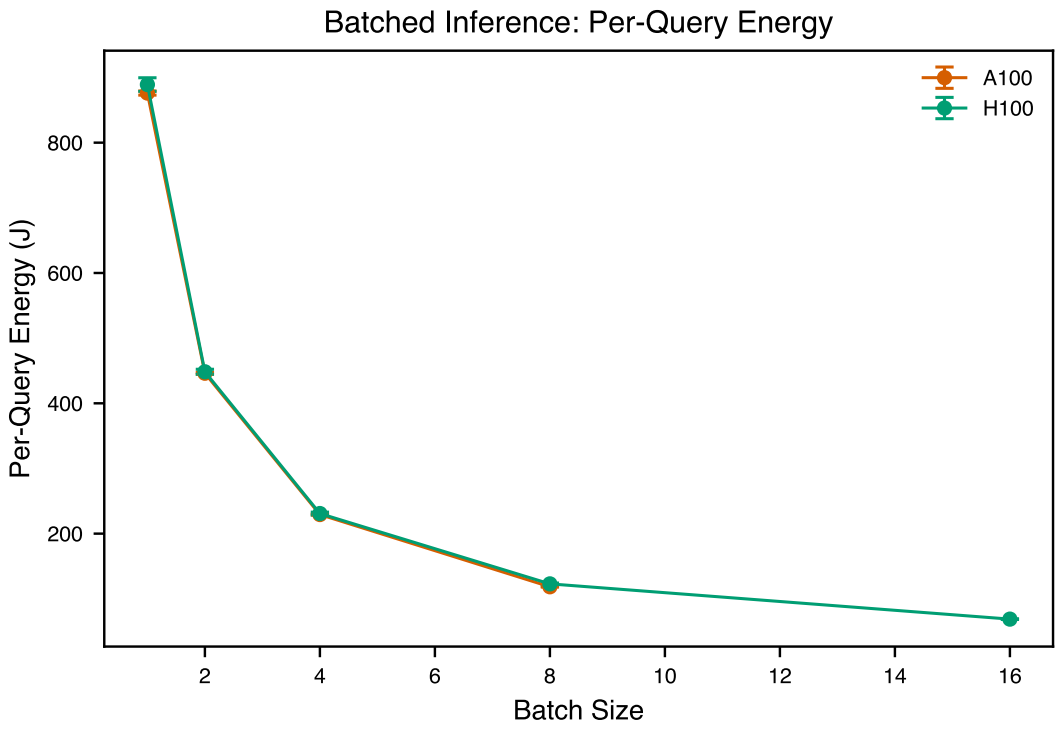

Figure 5: Batched inference: per-query energy. Per-query energy consumption as a function of batch size for text generation (Phi-3-mini-4k) on A100 and H100 GPUs. Batching amortizes the high fixed power cost of datacenter GPUs, reducing per-query energy. Error bars show ±1 standard deviation.

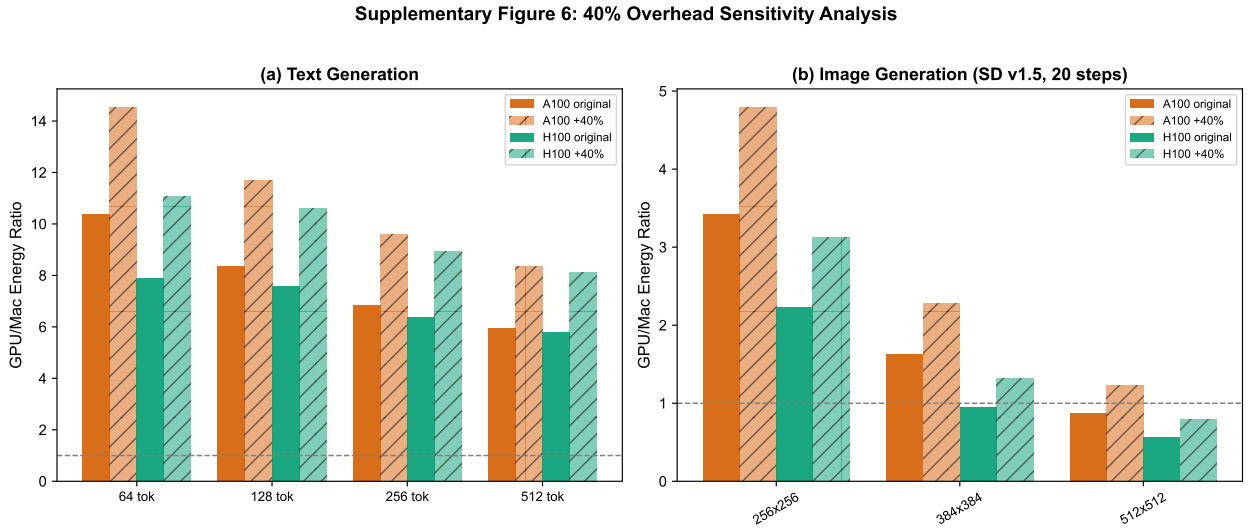

Figure 6: Sensitivity analysis: 40% overhead correction. Mac-to-GPU energy ratios before and after applying a 1.4× multiplier to GPU energy measurements to account for datacenter overhead (cooling, networking, storage). Even with this conservative correction that *reduces* the relative advantage of on-device computing, the crossover effect persists: local inference remains more efficient below the crossover point.

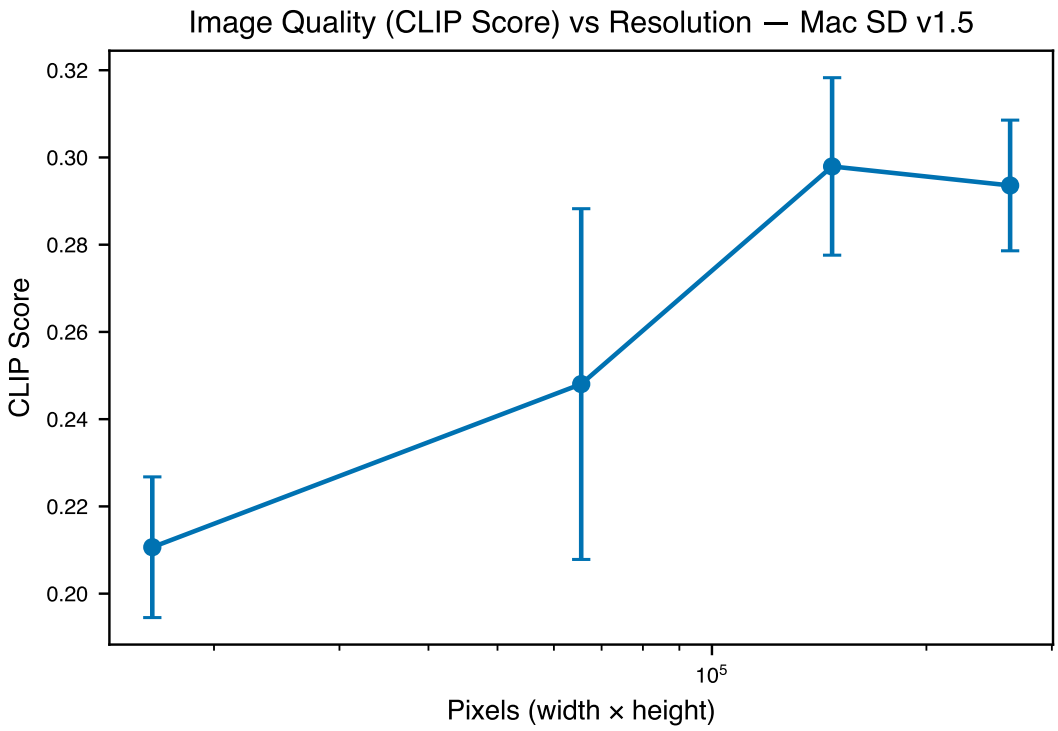

Figure 7: Image quality (CLIP score) vs resolution. CLIP similarity scores for SD v1.5 generated images as a function of resolution. Quality saturates at moderate resolutions, suggesting that generating at unnecessarily high resolutions wastes energy without meaningful quality improvement.

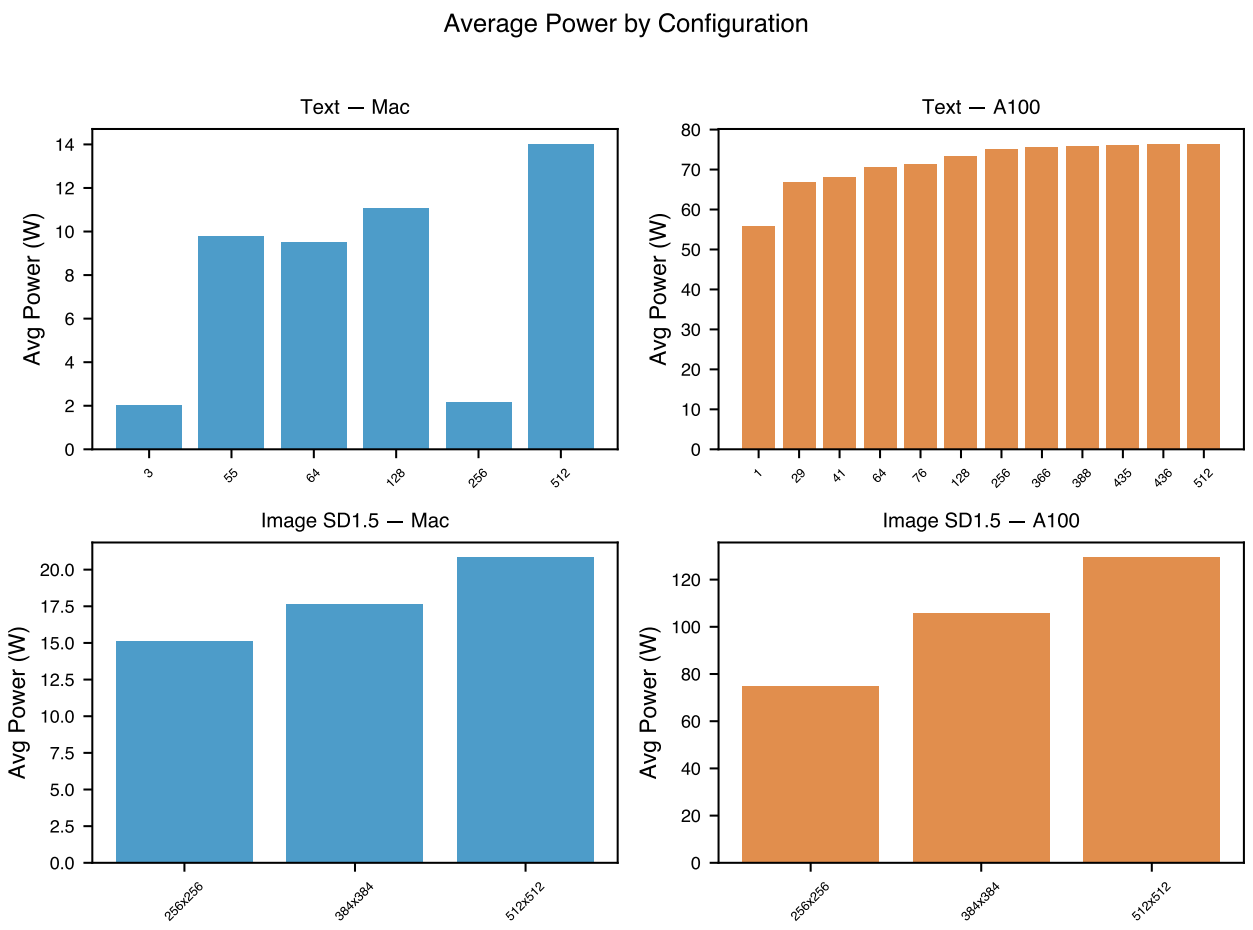

Figure 8: Average power consumption by configuration. Mean power draw (W) for each hardware platform across different generation configurations. Mac devices maintain low power (∼10–30 W) across all workloads, while datacenter GPUs draw 70–170 W regardless of task complexity, explaining why they are less efficient for small tasks.

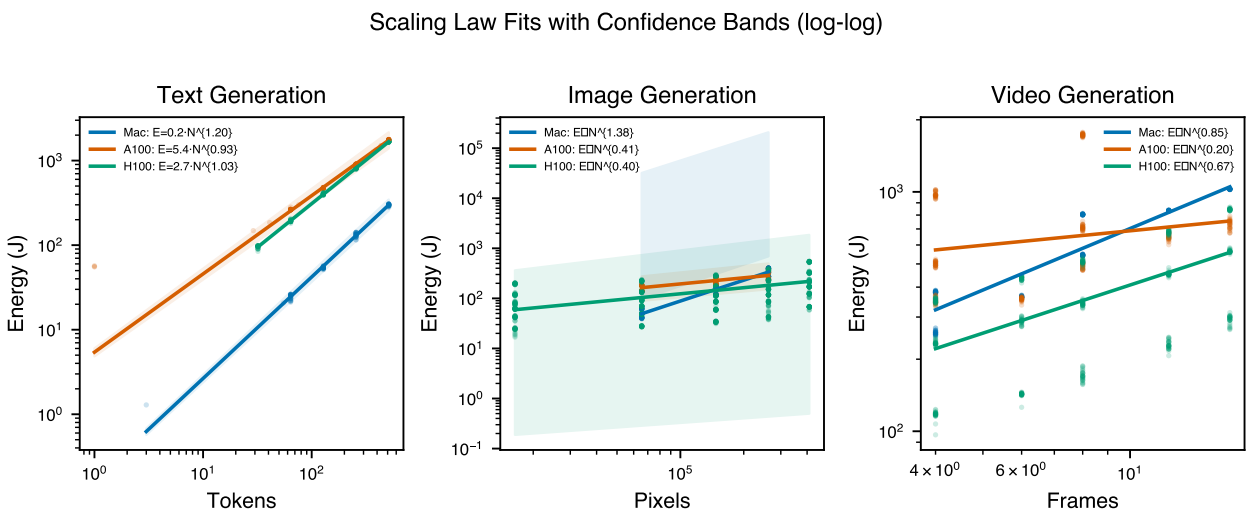

Figure 9: Scaling law fits with confidence bands. Log-log plots of energy (E) vs output size (N) for text (tokens), image (pixels), and video (frames) generation. Lines show power-law fits $E = a \cdot N^b$; shaded regions indicate 1σ confidence bands from parameter uncertainty. The Mac exhibits superlinear scaling ($b > 1$) across all modalities, while GPUs show sublinear scaling ($b < 1$) for diffusion workloads (image, video) and near-linear scaling for autoregressive workloads (text), reflecting different hardware utilization regimes.

2 Supplementary Tables

2.1 Supplementary Table 1: Per-Configuration Energy Statistics

Table 1: Complete per-configuration energy statistics. Mean, standard deviation, minimum, maximum, and number of measurements (n) for total energy consumption (J) across all experimental configurations and hardware platforms.

Hardware	Configuration	Mean (J)	Std (J)	Min (J)	Max (J)	n
Text						
Mac	128 tokens	53.2	6.3	20.8	57.2	30
Mac	256 tokens	127.5	24.9	1.3	142.6	30
Mac	3 tokens	1.3	0.0	1.3	1.3	1
Mac	512 tokens	288.2	54.9	0.0	307.6	30
Mac	55 tokens	20.8	0.0	20.8	20.8	1
Mac	64 tokens	24.5	1.6	21.2	26.3	30
A100	1 tokens	56.1	0.7	55.4	57.0	3
A100	128 tokens	443.9	98.7	57.0	482.6	30
A100	256 tokens	873.9	154.6	55.8	910.2	30
A100	29 tokens	148.5	0.0	148.5	148.5	1
A100	366 tokens	1266.2	0.0	1266.2	1266.2	1
A100	388 tokens	1340.5	0.0	1340.5	1340.5	1
A100	41 tokens	185.2	0.0	185.2	185.2	1
A100	435 tokens	1502.2	0.0	1502.2	1502.2	1
A100	436 tokens	1506.0	0.0	1506.0	1506.0	1
A100	512 tokens	1715.1	130.3	1266.2	1779.4	30
A100	64 tokens	254.2	40.4	55.4	282.0	30
A100	76 tokens	301.2	0.0	301.2	301.2	1
H100	128 tokens	402.3	7.9	388.9	417.2	30
H100	256 tokens	814.7	12.8	785.8	837.7	30
H100	32 tokens	95.4	3.5	84.4	99.1	30
... 2 more rows ...						
Image SD1.5						
Mac	256x256, 20 steps	41.2	1.3	38.9	44.6	30
Mac	256x256, 30 steps	62.5	2.2	55.9	67.3	30
Mac	384x384, 20 steps	120.2	4.3	113.0	137.2	30
Mac	384x384, 30 steps	178.6	4.9	173.9	194.8	30
Mac	512x512, 20 steps	272.0	3.4	265.1	278.9	30
Mac	512x512, 30 steps	399.7	3.7	392.5	407.0	30
A100	256x256, 20 steps	141.0	2.4	136.5	145.6	30
A100	256x256, 30 steps	182.2	3.4	174.1	188.5	30
A100	384x384, 20 steps	196.0	5.2	185.6	210.2	30
A100	384x384, 30 steps	267.0	6.5	253.0	277.2	30
A100	512x512, 20 steps	238.3	6.8	223.8	249.3	30
A100	512x512, 30 steps	333.1	8.7	320.5	356.3	30
H100	128x128, 10 steps	42.1	2.5	35.3	45.2	30

Hardware	Configuration	Mean (J)	Std (J)	Min (J)	Max (J)	n
H100	128x128, 15 steps	62.0	1.1	60.1	64.7	30
H100	128x128, 20 steps	79.3	4.5	70.1	89.3	30
H100	128x128, 30 steps	119.0	4.1	111.8	129.2	30
H100	128x128, 5 steps	23.6	2.4	16.8	26.0	30
H100	128x128, 50 steps	196.2	5.0	187.2	203.8	30
H100	256x256, 10 steps	48.9	0.8	47.2	50.4	30
H100	256x256, 15 steps	69.5	2.3	58.1	72.1	30
H100	256x256, 20 steps	92.0	2.4	88.7	99.9	30
... 21 more rows ...						
SDXL						
Mac	1024x1024	1531.5	12.0	1505.7	1551.2	30
Mac	512x512	315.8	2.6	309.9	320.5	30
Mac	768x768	768.6	3.8	761.2	776.9	30
A100	1024x1024	886.5	14.5	848.6	911.8	30
A100	512x512	319.5	10.0	295.3	343.9	30
A100	768x768	552.0	13.6	524.7	583.2	30
H100	1024x1024	525.2	257.0	177.3	968.0	150
H100	512x512	205.0	104.8	52.6	392.4	150
H100	640x640	265.6	135.6	83.1	499.0	150
H100	768x768	334.9	169.4	101.6	633.3	150
Video						
Mac	12 frames	834.8	2.7	829.2	843.3	30
Mac	4 frames	319.8	62.0	249.1	388.1	60
Mac	8 frames	674.4	130.4	538.5	814.1	60
A100	12 frames	648.0	18.2	601.4	678.1	30
A100	4 frames	609.0	266.3	330.7	1021.4	90
A100	8 frames	975.8	539.9	470.2	1773.5	90
H100	12 frames	454.0	183.4	206.9	705.2	90
H100	16 frames	566.9	225.4	265.9	863.4	90
H100	4 frames	234.3	95.8	96.6	364.6	90
H100	6 frames	288.8	119.8	125.9	449.6	90
H100	8 frames	340.8	139.1	156.7	535.7	90
Music						
Mac	11.22s	139.9	0.0	139.9	139.9	1
Mac	11.26s	138.1	0.0	138.1	138.1	1
Mac	11.31s	143.1	0.0	143.1	143.1	1
Mac	11.32s	141.8	0.3	141.5	142.2	2
Mac	11.44s	143.4	0.0	143.4	143.4	1
Mac	11.45s	144.0	0.2	143.8	144.3	2
Mac	11.4s	143.0	0.0	143.0	143.0	2
Mac	11.56s	144.7	0.0	144.7	144.7	1
Mac	11.58s	140.7	0.0	140.7	140.7	1
Mac	11.5s	143.5	0.0	143.5	143.5	1
Mac	11.61s	145.2	0.0	145.2	145.2	1

Hardware	Configuration	Mean (J)	Std (J)	Min (J)	Max (J)	n
Mac	11.63s	143.3	0.0	143.3	143.3	1
Mac	11.65s	145.8	0.0	145.8	145.8	1
Mac	11.69s	144.4	0.0	144.4	144.4	1
Mac	11.6s	143.8	0.0	143.8	143.8	1
Mac	11.71s	145.7	0.0	145.7	145.7	1
Mac	11.72s	143.2	0.0	143.2	143.2	1
Mac	11.79s	144.0	0.0	144.0	144.0	1
Mac	11.81s	143.2	0.0	143.2	143.2	1
Mac	11.83s	147.1	0.0	147.1	147.1	1
Mac	11.86s	143.2	0.0	143.2	143.2	1
… 105 more rows …						
Batched						
A100	batch=1	876.1	3.2	866.7	881.9	30
A100	batch=2	892.3	2.8	888.4	897.9	30
A100	batch=4	917.2	3.6	909.8	924.5	30
A100	batch=8	947.6	4.1	939.4	956.7	30
H100	batch=1	889.2	10.4	863.9	910.2	30
H100	batch=16	1101.0	8.7	1084.3	1120.9	30
H100	batch=2	896.7	7.7	885.6	921.5	30
H100	batch=4	923.4	8.6	909.7	951.6	30
H100	batch=8	983.3	10.2	964.3	1005.4	30

2.2 Supplementary Table 2: Detailed Crossover Analysis

Table 2: Energy crossover points. For each modality and platform comparison, we identify adjacent configurations where the Mac-to-GPU energy ratio crosses 1.0 (i.e., where on-device efficiency transitions relative to cloud). Energies are mean values in Joules; the crossover point is linearly interpolated.

Modality	Comparison	$\text{Config}_{\text{below}}$	E (Mac/GPU)	$\text{Config}_{\text{above}}$	E (Mac/GPU)	Crossover
Image SD1.5	Mac vs A100	('384x384', 30)	178.6/267.0	('512x512', 30)	399.7/333.1	218934

2.3 Supplementary Table 3: Sensitivity Analysis Results

Table 3: Sensitivity analysis: original vs 40%-corrected energy ratios. GPU energy is multiplied by 1.4× to account for datacenter overhead. The corrected ratio shows that even with this conservative adjustment, on-device inference remains more efficient for small-to-moderate workloads.

Modality	Comparison	Config	Mac (J)	GPU (J)	GPU_{corr} (J)	Ratio	$\text{Ratio}_{\text{corr}}$
Text	Mac vs A100	64	24.5	254.2	355.9	0.096	0.069
Text	Mac vs A100	128	53.2	443.9	621.5	0.120	0.086
Text	Mac vs A100	256	127.5	873.9	1223.5	0.146	0.104
Text	Mac vs A100	512	288.2	1715.1	2401.1	0.168	0.120
Text	Mac vs H100	64	24.5	193.4	270.8	0.127	0.091
Text	Mac vs H100	128	53.2	402.3	563.2	0.132	0.094
Text	Mac vs H100	256	127.5	814.7	1140.6	0.156	0.112
Text	Mac vs H100	512	288.2	1671.2	2339.7	0.173	0.123
Image SD1.5	Mac vs A100	256x256	51.9	161.6	226.3	0.321	0.229
Image SD1.5	Mac vs A100	384x384	149.4	231.5	324.1	0.645	0.461
Image SD1.5	Mac vs A100	512x512	335.9	285.7	399.9	1.176	0.840

3 Sensitivity Analysis

3.1 Methodology

Our primary energy measurements capture GPU-only power consumption via hardware-level power sensors (NVIDIA's `nvidia-smi` for datacenter GPUs and Apple's `powermetrics` for M-series chips). However, datacenter GPU deployments incur additional overhead from:

- **Cooling infrastructure**: Data centers typically operate at a Power Usage Effectiveness (PUE) of 1.1–1.4, meaning 10–40% additional energy is consumed for cooling.
- **Networking**: Request routing, load balancing, and data transfer consume additional power.
- **Storage and memory hierarchy**: Enterprise storage systems and memory management add to the energy budget.
- **Redundancy**: Enterprise deployments maintain redundant power supplies and backup systems.

To conservatively test the robustness of our crossover findings, we apply a 40% overhead multiplier (1.4×) to all GPU energy measurements. This represents the upper end of typical PUE values and does *not* include networking or storage overhead, making it a conservative correction that *favors* the GPU side.

3.2 Results

As shown in Supplementary Figure 6 and Supplementary Table 3, applying the 1.4× correction:

1. **Reduces** the Mac-to-GPU energy ratio by a factor of 1.4 across all configurations.
2. **Shifts** the crossover point to somewhat smaller workloads but does **not eliminate** the crossover effect.
3. Confirms that the fundamental phenomenon—on-device inference being more efficient for small-to-moderate workloads—is robust to reasonable estimates of datacenter overhead.

The persistence of the crossover under this correction strengthens our main finding: the energy advantage of on-device inference for small tasks is a structural property of the hardware power characteristics, not an artifact of measurement methodology.

4 Detailed Scaling Law Fits

We model the energy–output relationship as a power law:

$$E = a \cdot N^b \tag{1}$$

where E is total energy (J), N is the output size (tokens for text, pixels for images, frames for video), a is a platform-dependent constant reflecting base power draw and computational overhead, and b is the scaling exponent.

4.1 Fit Parameters

Table 4: Power-law fit parameters $E = a \cdot N^b$ with 1σ uncertainties from nonlinear least-squares regression.

Configuration	a ($\pm\sigma_a$)	b ($\pm\sigma_b$)
Text, Mac	0.1672 ± 0.0124	1.200 ± 0.012
Text, A100	5.4228 ± 0.5825	0.926 ± 0.018
Text, H100	2.6604 ± 0.0456	1.033 ± 0.003
Image, Mac	0.0000 ± 0.0000	1.382 ± 0.032
Image, A100	1.8277 ± 0.5243	0.405 ± 0.024
Image, H100	1.1719 ± 1.4802	0.405 ± 0.102

4.2 Interpretation

Key observations from the scaling analysis:

- **Mac platforms** consistently show $b > 1$ (superlinear scaling) across all modalities, indicating that energy costs grow faster than workload size. This is consistent with memory bandwidth saturation and thermal throttling at higher complexity on the unified memory architecture.
- **Datacenter GPUs** (A100, H100) show $b < 1$ (sublinear scaling) for diffusion workloads (image, video), reflecting improving GPU utilization as workload size increases—larger workloads better amortize the high idle power overhead. For autoregressive workloads (text), GPUs show $b \approx 1$ (near-linear scaling) because sequential generation cannot exploit additional parallelism.
- The **intercept** a is orders of magnitude smaller for Mac platforms, reflecting their fundamentally lower power envelope. This large difference in a is what creates the crossover: at small N, the low a of Mac dominates; at large N, the more favorable b of GPUs (from parallelism in diffusion workloads) compensates.
- For **image generation**, the exponent divergence is most extreme ($b_{\text{Mac}} = 1.395$ vs $b_{\text{A100}} = 0.374$), producing the sharpest crossover. The GPU's sublinear scaling reflects the diffusion model's ability to saturate GPU cores at higher resolutions.

These fits provide quantitative support for the energy crossover effect: the intersection of two power laws with different a and b parameters creates a natural crossover point whose location depends on the specific hardware and modality combination.

Supplementary Note 5: Prompt Diversity Analysis

To validate that energy consumption is determined by computational configuration rather than prompt content, we conducted a prompt diversity experiment using 5 semantically distinct prompts per modality (text and image), each repeated 10 times on the Mac M4 Pro platform.

Image generation shows near-complete prompt invariance: across 5 diverse prompts (landscape, cyberpunk, cat, astronaut, castle) at 512×512 resolution with 20 inference steps, the maximum difference between prompt means was only **2.0%** of the overall mean (207.8–211.9 J). The between-prompt variance (2.40) was 13× smaller than the within-prompt variance (18.15), confirming that run-to-run stochasticity dominates over prompt effects. This is expected because diffusion models execute an identical computational graph regardless of the text conditioning.

Text generation shows slightly higher prompt sensitivity (11.0% range across prompts at 128 tokens), driven by differences in input tokenization length (7–12 input tokens depending on prompt). However, the between-prompt variance (2.91) remained 14× smaller than the within-prompt variance (39.54), indicating that measurement noise still dominates. The prompt that produced early termination (38 tokens instead of 128) consumed proportionally less energy (14.6 J vs. ∼58 J), further confirming that token count, not prompt content, determines energy.

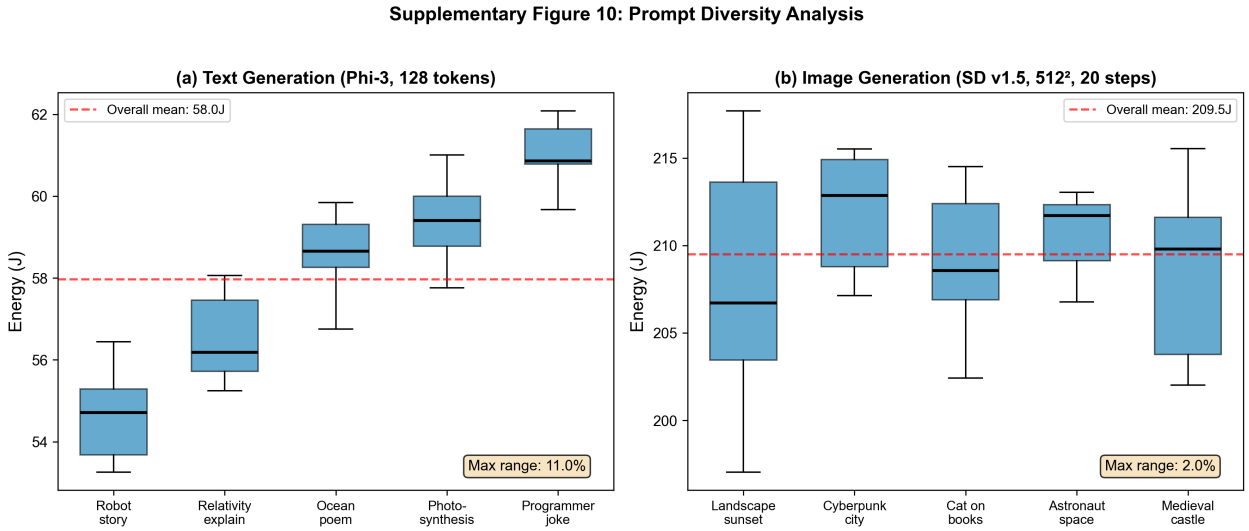

Figure 10: Supplementary Figure 10: Prompt diversity analysis. Box plots showing energy distribution across 5 semantically diverse prompts for (a) text generation (Phi-3, 128 tokens, 128-token runs only) and (b) image generation (SD v1.5, 512×512, 20 steps). Red dashed lines indicate overall means. Maximum between-prompt range is 11.0% for text and 2.0% for image, confirming that computational configuration, not prompt content, determines energy consumption.